

Place the centre of force S and let the body ride the curve at P . In the very next instant it sweeps the nascent arc PQ — the body's path in the least time, and this arc is the heart of everything that follows. Prop. I already told us that a central force makes the areas swept proportional to the times — but it never told us **HOW STRONG** the force is. The whole of celestial mechanics now waits on one question: can we read the magnitude of the force off the shape of the orbit alone?